

Ruthvik Uttarala

UI Software Engineer

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Work Experience

Software Engineer, AI & Cloud Architecture Feb 2026 - Present
OptimalHub LLC | Remote

- Built secure UniFi webhook intake in Firebase Cloud Functions on Google Cloud to flag unpaid entries, modeled for 500 events daily
- Implemented secure AWS S3 evidence delivery via signed URLs, sized for 750 violations monthly and 4500 image loads

Researcher Jun 2025 - Aug 2025
PlantVillage | University Park, PA, USA

- Fine tuned TensorFlow MobileNetV2 on 8000 augmented leaf images, delivering 83% accuracy and an F1 score of 0.81
- Accelerated React Native sync from 15s to 2s using Firebase Realtime Database and SQLite offline cache

Full Stack Developer, Idea Test Lab Program Jan 2024 - Mar 2024
Happy Valley Launchbox | State College, PA, USA

- Improved Node.js API response time 32%, from 1.9s to 1.3s, using MongoDB compound indexes and Express middleware
- Migrated REST endpoints from EC2 to AWS Lambda, lowering compute costs 25% while preserving throughput via concurrency tuning

Software Engineer, FastTrack Accelerator Program Sep 2023 - Dec 2023
Happy Valley Launchbox | State College, PA, USA

- Built a React ecommerce site using Next.js and TypeScript, cutting LCP 55% and raising Lighthouse mobile score from 50 to 70
- Secured Stripe payments with HMAC verified webhooks, processing 1200 monthly test transactions and handling retries idempotently using event dedupe

Projects

EdgeRAG Helpdesk Cloudflare Workers RAG Assistant 2026
Built edge RAG helpdesk for SRE runbooks, returning cited answers from tenant filtered Vectorize retrieval with topK 6

NovaArchitect Autonomous Infrastructure Strategy Agent 2026
Engineered a Next.js and FastAPI agentic cloud optimizer powered by Amazon Nova, enforcing structured JSON remediation plans with quantified risk scoring

Core Skills

- Programming Languages:** Python, JavaScript, TypeScript, C++, SQL, R, Java, C
- Backend & APIs:** Microservices architecture, REST APIs
- Front-End:** React, Next.js, jQuery, HTML5, CSS3
- Cloud and DevOps:** AWS, Azure, Tableau, Docker, Jenkins, Kubernetes, Firebase, CI/CD
- Engineering Practices:** Agile environment, Git, Troubleshooting, Version control
- Testing:** JUnit, Jest, Postman, Mockito, Test cases

Education

Pennsylvania State University - University Park Dec 2025
Bachelor of Science in Computer Science | Minor in Product Innovation Entrepreneurship